

honen’s *self-organizing map* concept. We utilize principles from quantum kernel approaches to identify the best matching unit on quantum computers. The design of our quantum neural network is directly motivated by the architectural framework of the classical self-organizing map (SOM).

Additionally, we are motivated by another relevant question concerning the representation of quantum states [38–40]. The pertinent issue is the depiction of states produced by quantum computers. Is there a more efficient method for visualizing quantum states with exceedingly large dimensions? One may contemplate augmenting the method to discern the topological structure intrinsic to quantum data, which may be produced from the simulations of quantum many-body systems. An unsupervised machine learning algorithm may assist in identifying distinct phases of such systems [41–46]. A low-dimensional map can be created that may correspond to different phases of these many-body systems. A similar study has already been performed with classical self-organizing maps [47]. In this context, we apply our method to the Schwinger model, a simplified representation of quantum electrodynamic interactions in $1 + 1$ -dimensional space-time.

The paper is organized as follows. In section II, we provide the necessary background information and outline our framework. We introduce the original version of Kohonen’s self-organizing map, along with its modified version designed to operate in kernel space. This section will also establish the notations used throughout the work. In section III, we introduce the proposed algorithm: the variational quantum self-organizing map or *Variational QSOM*. In section IV, we demonstrate the effectiveness of our algorithm on a classical data set (Fisher’s Iris data). Additionally, the phase space of the Schwinger model at $\theta = \pi$ is analyzed using the proposed algorithm. Finally, in section V, we discuss the theoretical aspects such as runtime complexity, robustness, and possible extensions of our algorithm.

II. BACKGROUND

In order to understand self-organizing maps (SOMs), it is helpful to provide a concise overview of the basic concepts of artificial neural networks as a whole, and then explain the unique features of SOMs within this context. There are three main types of biologically inspired artificial neural networks which are primarily studied in the deep learning literature: *signal-transfer networks*, *state-transfer networks*, and *competitive-learning networks* [13]. Layered feed-forward neural networks such as multilayer perceptrons and convolutional neural networks are some of the well-known examples of the signal-transfer networks. In these types of neural networks, the output signal has a unique dependence on the input signal. Learning in these networks is governed by the error-correcting back-propagation algorithms. State-

transfer networks, on the other hand, are based on relaxation effects. The strong nature of the feedbacks and non-linearities in these networks drives the internal states of the neurons to stable fixed points in the phase space. Hopfield networks and Boltzmann machines are exemplary networks from this category. The learning of these networks is governed by Hebbian learning or Boltzmann learning rules [48, 49]. As the name suggests, the learning of the networks in the third category is governed by competitive learning. The neurons in the output layer receive identical input information from the input layer, and they compete in their activities. Through lateral interactions, one of the neurons becomes the ‘winner’ and suppresses the activities of all the other neurons in the output layer. The winner neurons alternate depending on the received input. Once trained, different sections of the networks become sensitive to different parts of the vectorial input signals.

Based on such competitive learning rules, Kohonen introduced the idea of self-organizing maps inspired by the biological semantic or topographic maps created in the mammalian brain. SOMs have been used as a tool for unsupervised ML tasks, such as clustering, dimensionality reduction, and for learning a low-dimensional topology-preserving representation of the high-dimensional data sets.

A. Kohonen’s self-organizing map

Kohonen’s map or self-organizing map (SOM) is a type of competitive-learning network which is widely used for unsupervised learning. SOM is composed of two layers, an input layer and an output layer. The input layer feeds a data sample from the high-dimensional dataset $\mathbf{x} \in \Omega$, whereas the output layer consists of a lattice of neurons $\{l_1, \dots, l_k\} \in L$ arranged in a particular topology, for example, a rectangular or hexagonal grid. To each lattice point in the grid is associated a weight vector $\mathbf{w}_i \in \Omega$. The main objective of the self-organizing map is to learn the topology-preserving map $\kappa(\mathbf{x})$ from a high dimensional continuous space Ω (usually $\mathbb{R}^N$, where N corresponds to the dimensionality of $\mathbf{x}$) to a low-dimensional lattice space L of k neurons

$$\kappa(\mathbf{x}) : \Omega \rightarrow L. \quad (1)$$

The learning of this mapping, with the help of data, proceeds in three iterative steps as follows (see Fig. 1). In the first step, a sample $\mathbf{x}$ is selected randomly from the data. The second step comprises of finding the neuron l^* (i.e. l_{i^*}), termed *best matching unit (BMU)*, which is closest to $\mathbf{x}$. This is achieved by calculating the similarity score between $\mathbf{x}$ and the weight vector $\mathbf{w}_i$ associated with each neuron $l_i \in L$

$$\|\mathbf{x} - \mathbf{w}_{i^*}^t\| = \min_i \left\{ \|\mathbf{x} - \mathbf{w}_i^t\| \right\}, \quad (2)$$

where, superscript t indicates t^{th} iteration. Typically one uses Euclidean metric to calculate the similarity score;